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MPS/WPS-400 Series Programmable DC Power Supply User Manual

MATRIX TECHNOLOGY INC.

Applicable models: MPS-401,MPS-402,MPS-403,
MPS-404,MPS-405,MPS-406,
MPS-407,MPS-408,MPS-409,
MPS-410,MPS-411
WPS-401,WPS-402,WPS-403,
WPS-404,WPS-405,WPS-406,
WPS-407,WPS-408,WPS-409,
WPS-410,

Version number: V1.1

Preface

Dear users:

Hello! Thank you for purchasing brand-new MATRIX instruments and equipment. In order to use this instrument correctly, please read the full text of this manual carefully before using this instrument, especially the part about "safety precautions".

If you have read the full text of this manual, it is recommended that you keep it properly, and put it with the instrument or in a place where you can read it at any time for future reference.

Copyright information

- MATRIX TECHNOLOGY INC. all rights reserved.
- Products are protected by patents in China or other countries, including patents that have been obtained or are being applied for.
- MATRIX TECHNOLOGY INC. reserves the right to change product specifications and prices.
- It is the registered trademark of MATRIX TECHNOLOGY INC...

Verification and correction statement

Our company specifically declares that the instruments and equipment listed in this manual fully meet the specifications and characteristics specified in our technical specifications. This instrument has passed the factory calibration of our company before leaving the factory, and the calibration procedures and steps are in line with the specifications and standards of the electronic inspection center.

Product quality assurance

Our company guarantees that all the new instruments produced and manufactured have undergone strict quality confirmation, and at the same time, we guarantee that if any construction defects or parts of the products are found within one year after leaving the factory, our company will be responsible for repairing them free of charge. However, if the user changes the circuit and function by himself, or repairs the instrument and parts or damages the outer box, the company will not provide free warranty service. If all the ground wires are not properly connected according to the regulations or the machine is not operated according to the safety specifications, the company will not provide free warranty service.

This guarantee does not include accessories such as auxiliary equipment of this instrument that are not produced by our company.

During the one-year warranty period, please send the faulty unit back to our maintenance center or the dealer designated by our company, and our company will properly repair it.

If the unit breaks down under abnormal use, human negligence or beyond human control, such as earthquake, flood, riot, fire and other factors beyond human control, the company will not provide free warranty service.

(Our company follows the sustainable development strategy and reserves the right to improve the contents of this manual without prior notice.)

Safety precautions

The following general safety precautions must be observed in all stages of the operation of this instrument. Failure to follow these precautions or the specific warnings described in other parts of this manual will violate the safety standards for the design, manufacture and use of relevant instruments. Our company is not responsible for users' failure to observe these precautions.

Warning:

- Do not use damaged equipment, please check whether the equipment is in good condition before using it. Do not operate this equipment in an environment containing explosive gas, steam or dust.
- The power supply factory provides a three-core power cord, and your power supply should be connected to the three-core junction box. Before operating the power supply, you should first make sure that the power supply is well grounded to avoid accidental injury!
- Before connecting the device, please observe all the marks on the device.
- Please always use the cable provided when the instrument leaves the factory to connect the equipment to avoid accidental injury.
- When using wires with proper rated load, the capacity of all load wires must be able to withstand the maximum short-circuit output current of the power supply without overheating. If there are multiple loads, each pair of load wires must be able to safely carry the full rated short-circuit output current of the power supply.
- To reduce the risk of fire and electric shock, please ensure that the voltage fluctuation of the mains supply does not exceed 10% of the working voltage range.
- If you charge the battery with a power supply, make sure the positive and negative polarity of the battery when wiring, otherwise it will burn out the power supply!
- Do not use this equipment with the cover removed or loose.
- Do not install replacement parts on the instrument by yourself or make any unauthorized modifications.
- We are not responsible for direct or indirect financial losses that may occur when using this product.
- It is forbidden to use this equipment in life support systems or any other equipment with safety requirements.
- Failure to use the equipment in the manner specified by the manufacturer may undermine the protection provided by the equipment.
- Always clean the equipment housing with a dry cloth. Do not clean the inside of the instrument.
- The instrument needs to be monitored by personnel during operation, and the power supply should be turned off or disconnected when the instrument is not used for a long time.

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Chapter I Inspection and Installation

Please read the safety signs and instructions in this manual before installation or operation.

1.1 packing list

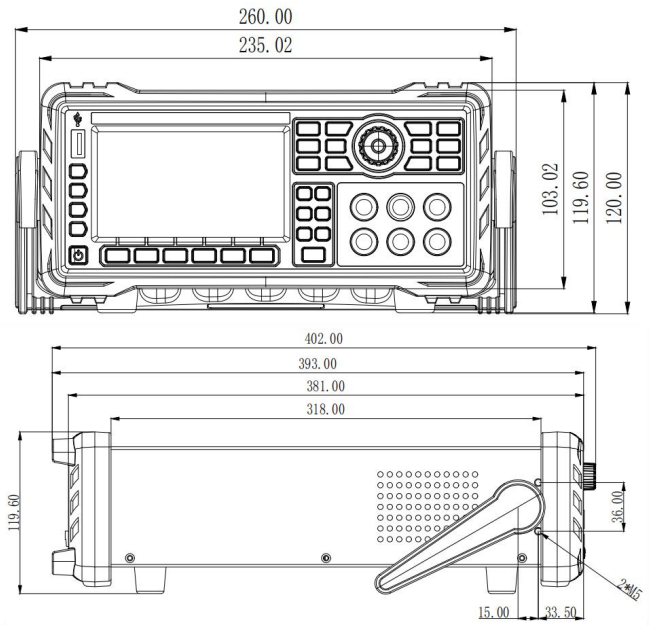
Open the package, please check the contents of the box before operating the instrument, and if there is any discrepancy, missing or appearance wear, please contact the seller immediately.

Item	Quantity	Remarks
Power supply host	1pc	
Power cord	1pc	Users can choose different power cords according to the specifications of power outlets in the region.
Test cable	1 pc	
Communication cable	1 pc	
User manua	1pc	
Product factory report	1pc	

After confirming that the packaging contents are consistent and there is no problem, please keep the packing box and related contents properly. The instrument needs to meet the packing requirements when returning to the factory for service.

1.2 Install the power supply

This instrument needs to be installed in a space with good ventilation environment and reasonable size. Please choose the appropriate space for installation according to the following power supply size introduction.
MPS-400/WPS-400 series size: 260mmW*120mmH*452mm (W: width; H: height; D: depth)
See the following dimension drawing for details:

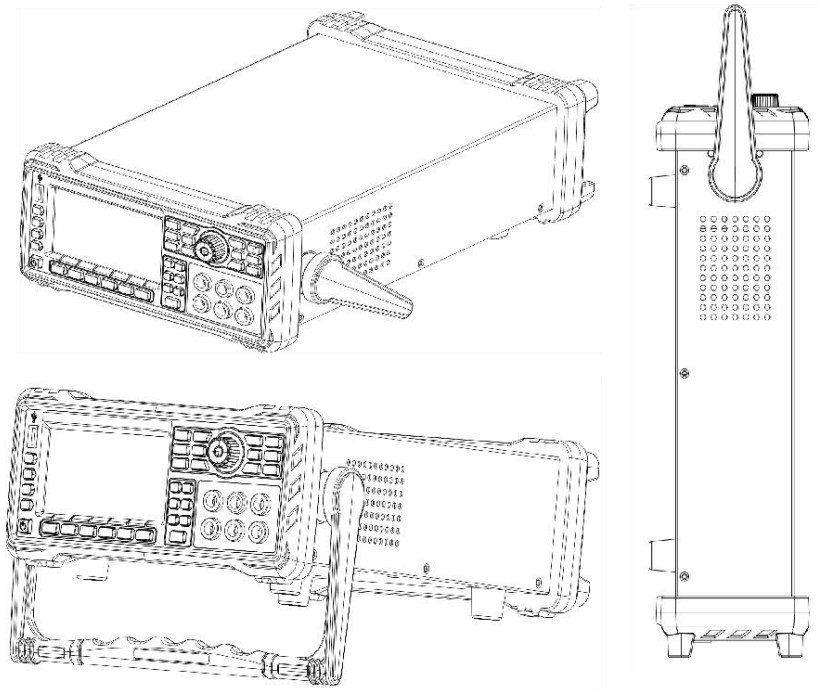


Unit: millimeter (mm)

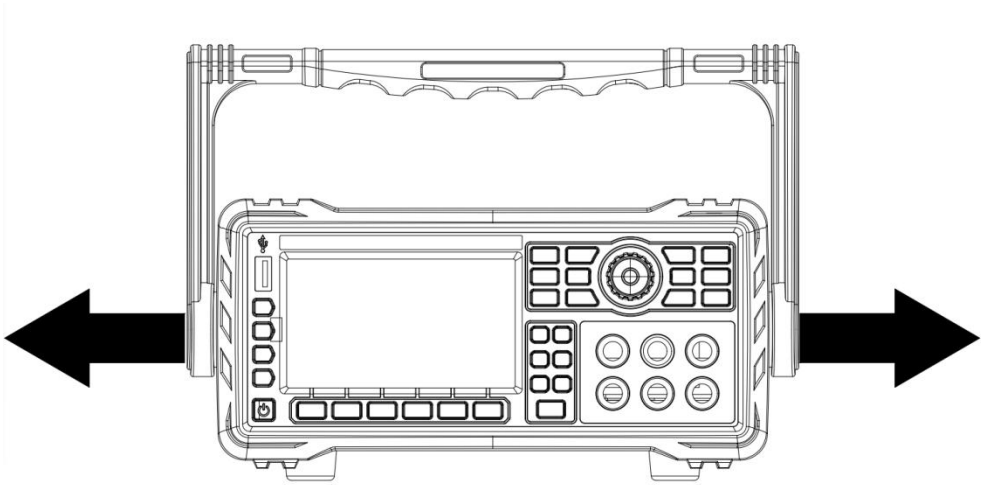
1.3 adjust the power handle

The power handle can be adjusted to the following three common states. Note that when adjusting the handle, you should pull your hands to both sides with moderate force and adjust the handle to the corresponding position.

Note: The following picture is for demonstration. Please refer to the actual appearance.



If it is necessary to remove the handle, you can adjust the handle to the position as shown below, and then pull it to both sides with both hands (as shown below).



Explanation: Do not apply too much force when loading and unloading the handle, be careful not to pinch your hands, and do not block the air inlet holes on both sides of the instrument or the exhaust holes on the rear panel.

1.4 Install the power cord

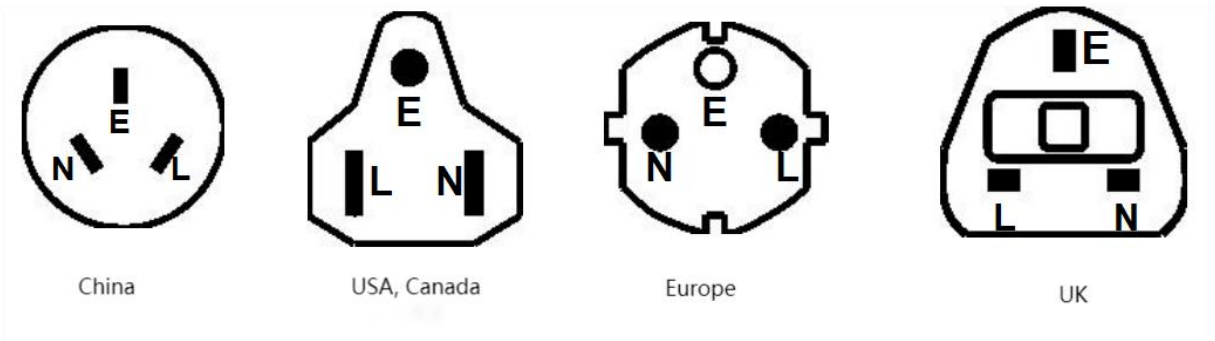
Connect the standard accessory power cord to ensure that the power supply has been supplied normally and grounded reliably.

Input requirements of power supply:
The working voltage of the power supply has two modes: 110V and 220V V. Please pay attention to the input voltage of the power supply. There is one item in the attachment for you.
Ground-matched power input line. If you find a mismatch, please contact the company's authorized dealer or after-sales service department immediately.

Input gear of AC power supply (which can be selected by the switch on the back of the power supply):

Option Opt.1: 220VAC ±10%47Hz-63Hz
Option Opt.2: 110VAC ±10%47Hz-63Hz

Type of power cord:
The model of power cord provided as standard is shown in the figure below. Please select the power cord model suitable for your local voltage from the power cord specification table below. If you make a mistake when buying the model, please contact the dealer or directly find the manufacturer to exchange it.



Chapter II Quick Start

This chapter briefly introduces the front panel, back panel, keyboard key function and TFT display function of MPS/WPS-400 series power supply, so as to ensure that you can quickly understand the appearance, structure and key function of the power supply before operating the power supply and help you use this series power supply better.

2.1 Introduction

MPS/WPS-400 series programmable DC power supply is a high-precision programmable linear DC power supply with single channel output.
MPS/WPS-400 series programmable DC power supply has clear operation interface, excellent performance index, and various communication interfaces can be selected to meet diverse test requirements.
The main special functions and advantages are as follows:

- 4.3-inch high visibility TFT display screen
- The voltage and current can be adjusted by using the numeric keyboard or knob.
- High accuracy and resolution, voltage resolution: 1mV, current resolution up to 1mA.
- List output function
- Battery function
- Low ripple and low noise
- Intelligent fan control saves energy and reduces noise.
- With CC/CV priority function.
- The rising slope of voltage and current can be set.
- With SCPI and MODBUS instructions, it is convenient to set up an intelligent test platform.
- It has the functions of over-voltage, over-current protection and over-temperature protection.
- Support interface display in Chinese and English.
- With voltage compensation function

Power specification model Tabel

MPS-400 series programmable DC power supply selection table:

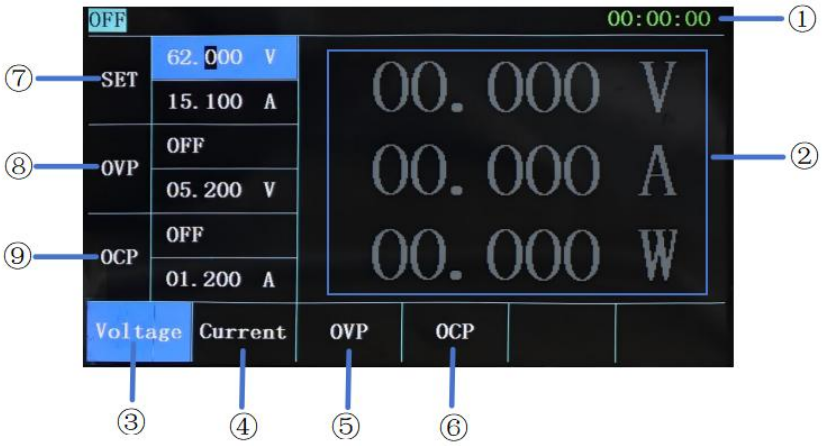
Model	Voltage	Current	Power
MPS-401	15V	60A	900W
MPS-402	20V	30A	600W
MPS-403	20V	45A	900W
MPS-404	30V	20A	600W
MPS-405	30V	30A	900W
MPS-406	60V	10A	600W
MPS-407	60V	15A	900W
MPS-408	80V	6A	480W
MPS-409	80V	11A	880W
MPS-410	150V	3A	450W
MPS-411	150V	6A	900W

WPS-400 series programmable DC power supply selection table:

Model	Voltage	Current	Power
WPS-401	30V	40A	400W
WPS-402	30V	60A	900W
WPS-403	40V	40A	400W
WPS-404	40V	40A	900W
WPS-405	60V	20A	400W
WPS-406	60V	30A	900W
WPS-407	80V	20A	400W
WPS-408	80V	20A	900W
WPS-409	150V	10A	400W
WPS-410	150V	15A	900W

2.2 Introduction of Front Panel

①USB interface (5V/1A output, which can be used for software upgrade and data import and export).
② ⑤ Function buttons
③ Power switch
④ Function keys corresponding to the menu at the bottom of the screen (menus in different display interfaces are different)
⑥⑦ Source output terminal (black is negative and red is positive)
⑧End voltage compensation terminal
⑨Pet-name ruby detachable handle
⑪Multifunctional knob
⑩⑫Attending number keys and "Enter" confirmation key.
⑬TFT display screen
⑭Power specification model label



① Power status display column (display output status and OVP/OCP/OTP/ remote compensation /PC connection /USB and output time, etc.)
②Power output voltage, current and power parameter display column.
③④⑤⑥Voltage, current and OVP/OCP setting instructions. When the corresponding position is lit, it means that the parameter can be set. Use the corresponding function key at the bottom of the screen to select the parameter to be modified.
⑦. Set values of voltage and current (refer to 3.3 Current setting).
⑧Over-voltage setting value and over-voltage enabling status (refer to 3.5.1 OVP setting).
⑨. Over-current setting value and over-current enabling status (refer to 3.5.2 OCP setting).

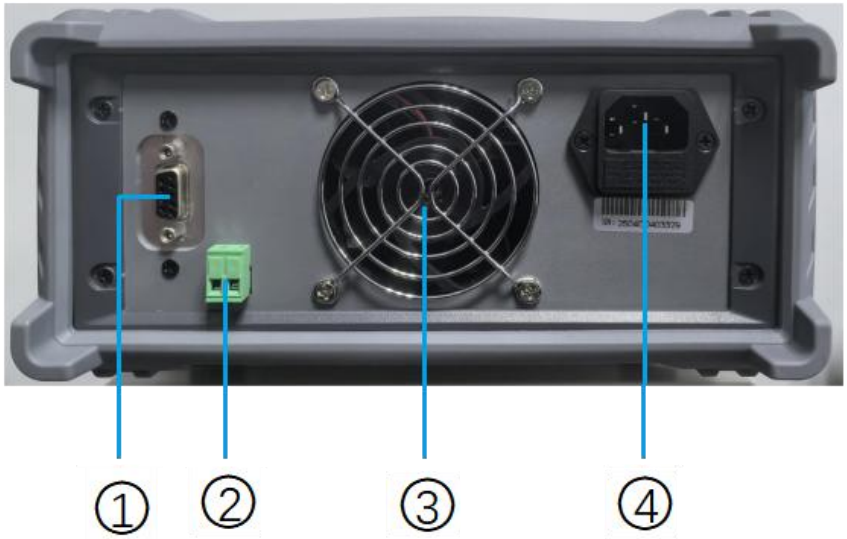
2.3 Introduction of Keyboard Keys

Keys are described in the following table:

button	Name and function
0-9	Digital input key
⊙	decimal point
Enter	Enter to confirm the entered parameters.
	Multi-function knob, used to set the relevant parameters of power supply.
Menu	Menu menu function key, used to set the relevant parameters of power supply.
Esc	Esc cancel/return key
Store	Store storage file
Local	Local local switch key to return the power supply from remote mode to local mode.
HotKey	HotKey hotkey, enter or exit hotkey function.
Lock	Lock panel keyboard lock
Output	The output on/off key is used to control the output state of the power supply.
Display	Display output waveform display button
Battery	Battery battery function setting key
List	List key, which is used to set the List parameter.
Ω	Ω resistance measuring function key
⏻	Power switch button

2.4 Introduction of Back Panel

Rear panel layout:

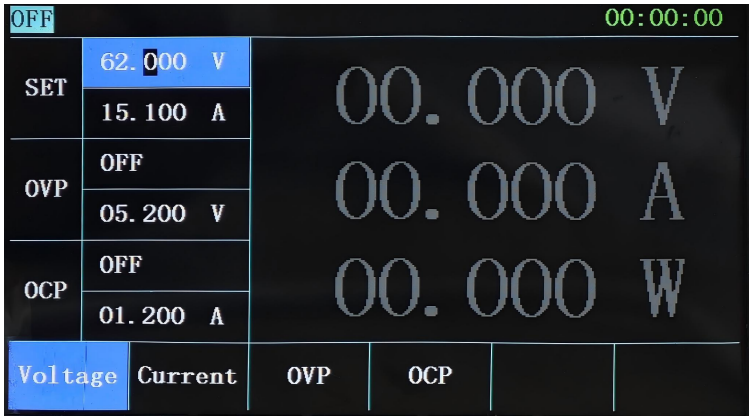


①Communication port (standard RS-232 communication port, optional USB, RS-485, LAN port).
②Voltage compensation port (S+: positive terminal of voltage compensation, S-: negative terminal of voltage compensation,).
③Heat dissipation vents.
④Power jack (AC power input interface).

2.5 POST

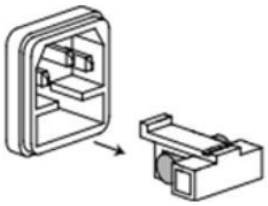
The successful self-inspection process shows that the power products purchased by users meet the factory standards and can be used by users normally. Before operating the power supply, please make sure that you have understood the safety instructions.
Please make sure that the power supply voltage is consistent with the power supply voltage before turning on the power supply, otherwise it will burn out the power supply.
Please connect the main power plug to the power socket with protective grounding, and do not use the wiring socket without protective grounding. Before operating the power supply, you should first make sure that the power supply is well grounded.
Please pay attention to the positive and negative signs before wiring the power supply, otherwise it will burn out the power supply.

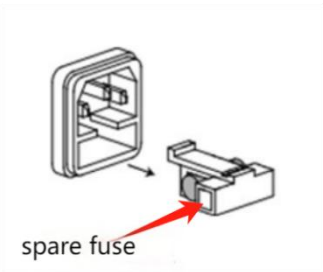
Self-inspection steps:
The normal self-check process of power supply is as follows:
1. Connect the power cord correctly, press the power switch to turn on the power supply, and the power supply performs self-inspection.
2. The self-check of power supply is completed, and the display status of TFT display screen is as follows.



Exception handling:
When starting the power supply, the power supply cannot be started normally. Please refer to the following steps to check and handle it.
1. Check whether the power cord is connected correctly, and confirm that the power supply is powered.
Power cord connected well => 2
Power connection error = > Please reconnect the power cord to see if the abnormality is cleared.
2. Whether the power supply is turned on. Press the power on key to be in the power on state.
Yes => 3
No = > Please press the voltage switch to turn on the power supply to see if the abnormality is cleared.

3. Check whether the fuse of the power supply is burnt out.
If the fuse is burnt out, please replace it. The specific steps are as follows:
1) Unplug the power cord and take out the fuse box with a small screwdriver. (See the introduction of the rear panel for fuse location)





2) Replace the fuse with the same specification, put it in the box and reinstall it.

Fuse specification list:

model	Fuse specification (220V)
MPS-408，MPS-410，WPS-401，WPS-403，WPS-405，WPS-407，WPS-409	5A
MPS-402，MPS-404，MPS-406	6A
MPS-401，MPS-403，MPS-405，MPS-407，MPS-409，MPS-411，WPS-402，WPS-404，WPS-406，WPS-408，WPS-410	8A

2.6 output check

The output inspection can ensure that the power supply reaches its rated output and can correctly perform the front panel operation.

Output voltage check:
Verify the basic voltage function of the power supply without load.
1. Turn on the power supply.
2. Set the power supply current value (⩾0.1A).
3. Turn on the power output.
Press the ON/OFF function button, and the ON and CV status signs on the TFT display will light up.
4. Set the power supply voltage.
Set different power supply voltages, and check whether the voltage value displayed on the TFT is close to the set voltage value and whether the current value displayed on the TFT is close to 0A.
5. Ensure that the power supply voltage can be adjusted from 0V to the maximum output voltage.
-End

Output current check:
Verify the basic current function of the power supply when the output is short-circuited.
1. Turn on the power supply.
2. Turn OFF the power output, make sure that the power supply is in the OFF state, and display the off state sign on the TFT.
3. Connect an insulated wire between the output terminals (+) and (-) of the power supply. The wire used must be able to withstand the maximum output current of the power supply.
4. Set the power supply voltage to 1V.
5. The power output is on.
6. Set the power supply current.
Set different power supply current, and whether the current value displayed on TFT is close to the set current value.
7. Ensure that the power supply current can be adjusted from 0A to the rated full output current of this channel.
8. Turn off the power output and remove the short-circuit conductor.
-End

Chapter III Functions and Characteristics

This chapter will describe the functions and characteristics of the power supply in detail. Will be divided into the following parts:

- Local (switch local/remote) operation
- Voltage setting operation
- Current setting operation
- Output on/off operation
- Overvoltage/overcurrent protection function
- Display (waveform display) function
- List operation
- Battery function
- Ω (resistance measurement) function
- Menu function
- Esc (cancel/exit)
- Store function
- Hol Key function
- Lock function

3.1 Switching local/remote operation

The power supply provides two operation modes: local operation and remote operation. The two operation modes can be switched through communication instructions. The power initialization mode defaults to the local operation mode.
Local operation mode: use the buttons on the front panel of the power supply for related operations.
Remote operation mode: the power supply is connected to the PC, and the related operation of the power supply is carried out through communication instructions on the PC. Power supply is far away
In the process operation mode, except for the Local key, other buttons on the panel do not work. You can switch to Local operation mode through local. When the operation mode changes, it will not affect the output parameters of the power supply.

3.2 voltage setting operation

The voltage setting range is 0V to the maximum output voltage value. When the function key corresponding to "Voltage" at the bottom of the screen is pressed, the voltage setting position of the display screen will be lit, and the voltage setting operation can be performed at this time. The output voltage value can be set through the front panel in the following two ways.
①After the power is turned on, press the function key corresponding to "voltage" at the bottom of the screen, move the cursor to the specified position by pressing the knob and then turn the knob to adjust the voltage setting value.
②After the power is turned on, press the function keys+0 to 9 corresponding to "Voltage" at the bottom of the screen and then press the Enter key to set the voltage value.

3.3 Current setting operation

The range of current setting is 0V to the maximum output voltage. When the function key corresponding to "Current" at the bottom of the screen is pressed, the current setting position of the display screen will be lit, and the current setting operation can be performed at this time. The output current value can be set through the front panel in the following two ways.

①After the power is turned on, press the function key corresponding to "Current" at the bottom of the screen, move the cursor to the specified position by pressing the knob and then turn the knob to adjust the current setting value.
②After the power is turned on, press the function key+0 to 9 corresponding to "Current" at the bottom of the screen and then press the Enter key to set the current value.

3.4 Output On/Off Operation

You can control the power supply output by pressing the Output key on the front panel.

When the power output is in the ON state, the working status signs (ON and CV/CC) in the status bar at the top of the TFT display screen will be lit, and the timer in the upper right corner will be cleared and start to beat. When the power output is in the OFF state, the working status sign (OFF) on the TFT will be lit, and the time in the upper right corner will stop beating to keep the current output timer.

explain

After the power supply is connected to the object to be tested, press the On/Off key to turn on the output, otherwise it is easy to strike a light when wiring. If the power supply has no output after the output is turned on, please check the voltage and current settings, and set both the voltage and current to non-zero values before turning on the output. When the power output is off, the voltage, current and power display windows are gray; when the power output is in CV state, the voltage display window is green, the current display window is yellow and the power display window is yellow. When the power output is in CC state, the voltage display window is yellow, the current display window is red and the power display window is yellow.

3.5 overvoltage/overcurrent protection function (OVP/OCP)

The overvoltage/overcurrent protection function allows the user to set an overvoltage/overcurrent protection point. When the voltage/current of the measured object is greater than this protection point, the power supply will carry out OVP/OCP protection. When the overvoltage current is protected, the power output will be turned off, accompanied by the buzzer, and the VFD will display the relevant alarm prompt (OVP/OCP).

3.5.1 OVP (overvoltage protection):

When the function key corresponding to "Overvoltage" at the bottom of the screen is pressed, the power supply will enter the overvoltage protection setting operation, and the overvoltage setting parameter at the overvoltage setting position of the display screen will be lit. At this time, when the function key corresponding to "Overvoltage" at the bottom of the screen is pressed again, it is possible to switch between the overvoltage protection parameter setting and the overvoltage protection switch state. When the overvoltage protection parameter is set, the parameters of overvoltage protection value can be modified by knob or 0-9 number keys (please refer to 3.2 Voltage Setting Operation for the method). When the overvoltage protection switch is set, turn the knob left and right to change the ON/OFF state of overvoltage protection.

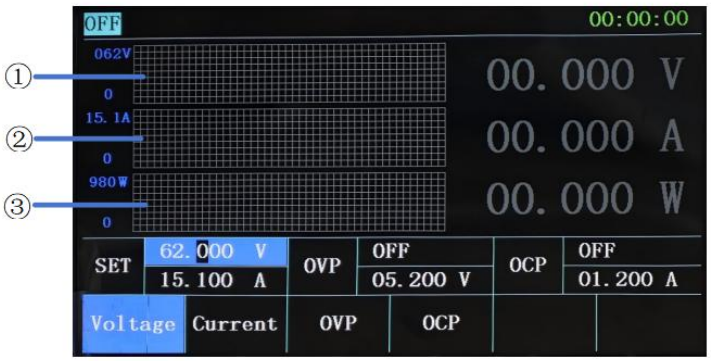
3.5.2 OCP (Overcurrent Protection):

When the function key corresponding to "overcurrent" at the bottom of the screen is pressed, the power supply will enter the overcurrent protection setting operation, and the overcurrent setting parameters at the overcurrent setting position of the display screen will be lit. At this time, when the function key corresponding to "overcurrent" at the bottom of the screen is pressed again, you can switch between the overcurrent protection parameter setting and the overcurrent protection switch state. When the overcurrent protection parameters are set, you can modify the parameters of the overcurrent protection

value with the knob or the number keys 0-9 (please refer to 3.3 Current setting operation for the method). When the overcurrent protection switch is set, turn the knob left and right to change the ON/OFF state of the overvoltage protection.

3.6 Display (waveform display) function

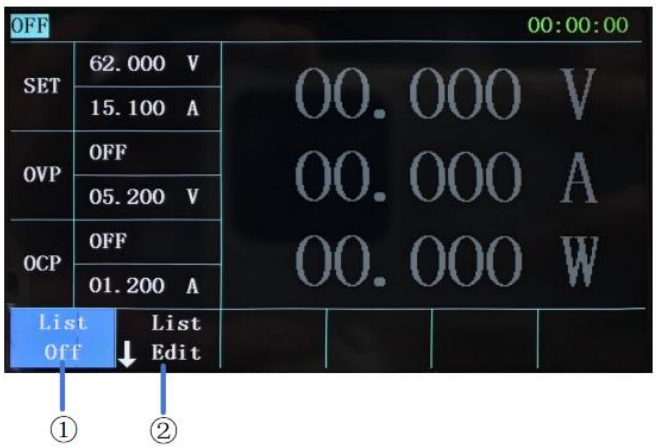
Display function can monitor the waveforms of voltage, current and power. Press Display in standby mode to enter the waveform display interface (as shown below), and press Display again to exit the waveform display function.



①Output voltage waveform
②Output current waveform
③Output power waveform

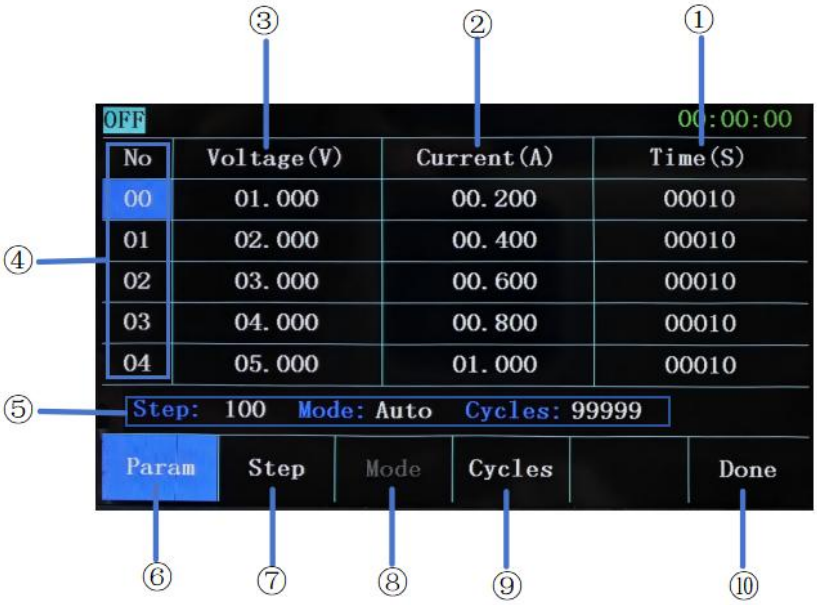
3.7 List operation function

The List operation function can set the voltage, current and time of each step by editing the List file. When the power supply receives a trigger signal, the power supply can automatically execute the set value of each step. There are three trigger signal modes: automatic, key and external.



①Sequence on/off: the List operation function switch is turned on and off by pressing the function key corresponding to "sequence on/off" at the bottom of the screen. After the List operation function is turned on, only the Output key on the front panel is effective, and other key operations can be performed only after the List operation function is turned off.
②. Sequence Editing: Set the function parameters of List operation, and enter the function parameters of List operation by pressing the function key corresponding to "Sequence Editing" at the bottom of the screen.

3. 7.1 List operation function parameter setting.



①. time: the stay time of this step for List operation (0-99999 seconds).
②. current: List (sequence) operation The maximum output current of this step.
③. voltage: List (sequence) operation The output voltage of this step.
④Step: List (sequence) operation current step (step 0-99).
⑤. Steps: the total steps (0-100 steps) of the current List operation.

Mode: At present, the trigger signal mode of List operation can be divided into three types: automatic, key press and external.

Loop: the number of loops executed by the current List operation (0-99999, 0 is an infinite loop).

⑥ Parameters: This option is used to set the time, current, voltage and other parameters of the List operation. Press the function key corresponding to "Parameters" at the bottom of the screen to enter parameter setting. Turn the knob left and right to select the steps to be edited. Press the knob to move the cursor to select the parameters such as time, current and voltage to be edited. You can modify the parameters such as time, current and voltage by turning the knob left and right or the number keys and the confirmation key.

⑦. Steps: this option is used to set the general steps of the List operation. Press the function key corresponding to "Steps" at the bottom of the screen to enter the step number setting. After selection, you can modify the step number by turning the knob or the number keys left and right.

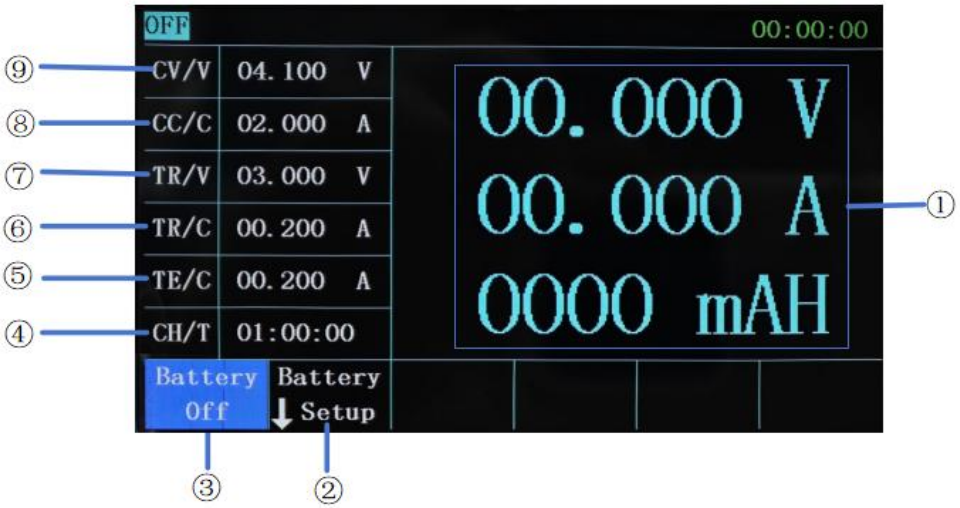
⑧. Mode: This option is used to set the trigger mode of List operation. You can enter the mode setting by pressing the function key corresponding to "Mode" at the bottom of the screen, and you can modify it by turning the knob left and right after selection. The trigger signal mode is divided into three modes: automatic, key press and external.

⑨Cycle: this option is used to set the cycle number of List operation (0-99999, 0 is infinite cycle). Press the function key corresponding to "Cycle" at the bottom of the screen to enter the cycle number setting, and after selection, you can modify it by turning the knob left or right or the number keys and the Enter key.

⑩Finish: After setting the List operation parameters, press the function key corresponding to "Finish" at the bottom of the screen to complete the setting of the List operation parameters and return to the main interface of the List operation function.

3.8 Battery function,

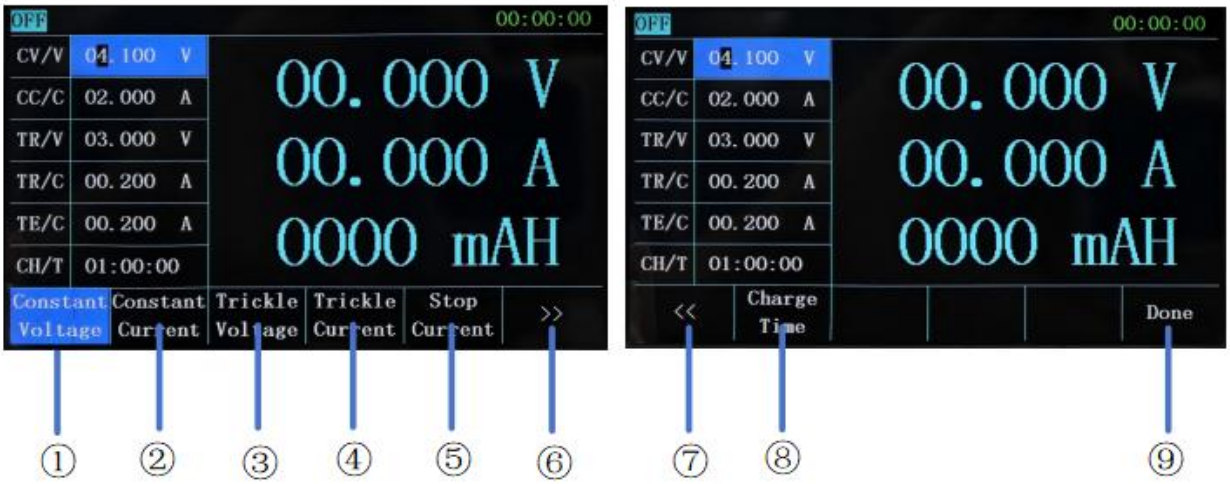
The Battery function can simulate the charger to charge the battery.



①. Display columns for parameters such as power output voltage, current and capacity.
②. Battery setting, enter the battery charging parameter setting function by pressing the function key corresponding to "Battery setting" at the bottom of the screen.
③. Battery function switch, the Battery function is turned on and off by pressing the function key corresponding to "Battery On/Off" at the bottom of the screen. After the battery function is turned on, the battery can be charged by pressing the Output key on the front panel. At this time, the battery capacity (mAH position) in the parameter display column will increase continuously, and the output will be automatically turned off after reaching the predetermined charging parameters, and the charging will be terminated. After the Battery function is turned on, only the Output key on the front panel is valid, and after the Battery function is turned off, other key operations can be performed.
④. Time, charging time setting (up to 99: 59: 59), the power supply will automatically stop charging when the charging time is up.
⑤The final current and charging termination current are set. When the charging current is less than this set current, the power supply will automatically stop charging after 5 seconds.
⑥Trickle, when the battery voltage is lower than the set value of trickle, the power supply will charge the battery according to the current value set by trickle, which can protect the battery.
⑦Trickle, Trickle and Standard Flow charging thresholds. When the battery voltage is lower than the set value of Trickle, the power supply will charge the battery according to the current value set by Trickle, and when the battery voltage is higher than the set value of Trickle, the power supply will charge the battery according to the current value set by Standard Flow.
⑧Standard current, when the battery voltage is higher than the set value of trickle voltage, the power supply will charge the battery according to the current value set by standard current.
⑨Standard voltage, the voltage value of battery charging.

3.8.1 Battery Settings

Press the function key corresponding to "Battery Settings" at the bottom of the screen to select the power supply to enter the battery charging parameter setting function. After entering the battery parameter setting function, select the parameters to be set by pressing the corresponding function key at the bottom of the screen, turn the knob left and right to modify the parameters, press the knob to move the cursor of the parameter setting position, or directly enter through the numeric keys and press Enter to confirm.

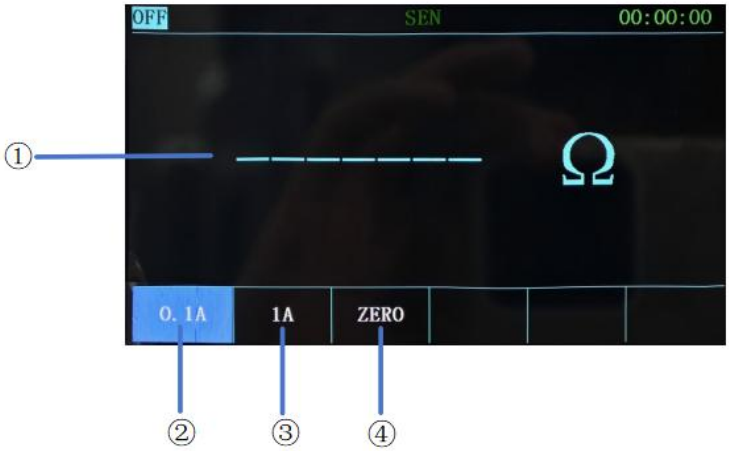


①Standard voltage (standard voltage), the voltage value of battery charging.
②Standard current (standard current), when the battery voltage is higher than the trickle setting value, the power supply will charge the battery according to the current value set by standard current.
③Trickle voltage (trickle voltage), the threshold of trickle and standard current charging. When the battery voltage is lower than the set value of trickle voltage, the power supply will charge the battery according to the set value of trickle voltage, and when the battery voltage is higher than the set value of trickle voltage, the power supply will charge the battery according to the set value of standard current.
④Trickle current (trickle), when the battery voltage is lower than the trickle setting value, the power supply will charge the battery according to the current value set by trickle, which can protect the battery.
⑤Termination current (final current), charging termination current setting. When the charging current is less than this setting current, the power supply will automatically stop charging after 5 seconds.
⑥Next page is used to switch to the setting interface of the next page.
⑦Previous page, used to switch to the setting interface of the previous page.
⑧Charging time (time), charging time setting (up to 99: 59: 59), and the power supply will automatically stop charging when the charging time is up.
⑨Finish. After the charging parameters are set, press the function key corresponding to "Finish" at the bottom of the screen to complete the charging parameters setting.

3.9 Ω (low resistance measurement) function

Press the key to enter the "Ω" low resistance measurement function (the screen is shown below), and switch gears by pressing the corresponding gear key at the bottom of the screen. The "1A" gear is recommended for low resistance, and the measurement range of low resistance is from 0.0001Ω to 200Ω.

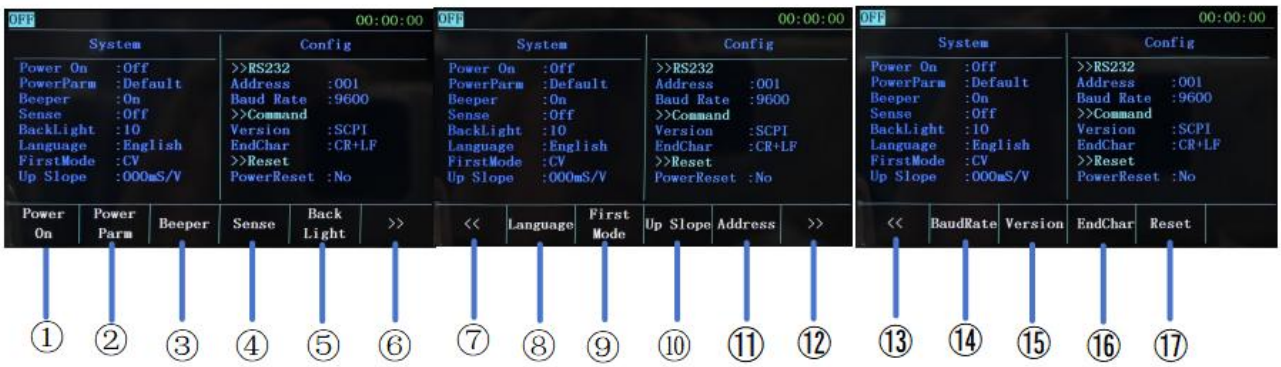
Press key again to exit the "Ω" resistance measurement function.
① Resistance value display window.
② 0.1A measuring file.
③ 1A measuring gear ("1A" gear is recommended for low resistance).
④ Clear the function key (this function can only take effect if the test line is short-circuited).



Note: 4-wire measurement and SENSE (voltage compensation interface must be wired) must be used when using the resistance measurement function.

3.10 Menu function

When the power supply is in standby mode, press the MENU key to enter the Menu function, and you can set the relevant parameters of the power supply. After entering the Menu function, switch to the Menu to be edited by pressing the corresponding function key at the bottom of the screen. After selecting the menu to be edited, modify the selected menu parameters through the corresponding function key at the bottom of the screen, press the "Back" function key to return to the main menu after modification, and press the



"Menu" or "ESC" key again after all parameters are modified to save and exit the menu function.

3.10.1 System parameters

① Power-on state: the power-on output state remains set, "On" means to keep the power-on output state, "Off" means to keep the power-on output state closed, and "Last time" means to keep the power-on output state before the last shutdown.

② Power-on parameters: Power-on parameters remain set, "Last" means to keep the data when the power was cut off last time, and "Default" means to keep the factory default value.

③ Buzzer: Power button sound setting, "On" means to turn on the buzzer sound, and "Off" means to turn off the buzzer sound.

④ Remote compensation: the remote compensation function of the power supply is set. ON means that the remote compensation function is turned on, and OFF means that the remote compensation function is turned off.

⑤Brightness: the display brightness level of the power TFT display screen can be adjusted between 1 and 10, with 1 being the darkest and 10 being the brightest. Turn the knob left and right to adjust, and exit the background brightness setting by the corresponding function key "Done" at the bottom of the screen.

⑥ >>: Turn the page to the right.

⑦ >> Turn the page to the left.

⑧Language: The language settings of the power operating system are available in Chinese and English.

⑨Priority mode: the priority mode is set when the power output is turned on. There are voltage priority and current priority to choose from. When the voltage is priority, the voltage is turned on first when the power output is turned on, and when the current is priority, the current is turned on first.

⑩ Ascending slope: set the climbing speed when the power output is turned on. You can set the climbing speed of voltage or current, and choose one of them. The unit is mS/A or mS/V, and the range is 0-250. Turn the knob left and right to adjust, and exit the setting of ascending slope through the corresponding function key "Done" at the bottom of the screen.

⑫ >>: Turn the page to the right.

⑬ >>: Turn to the left.

3.10.2 Configuration parameters

Serial port: communication port setting.

⑪Address: power communication address setting, ranging from 1 to 254. Turn the knob left and right to adjust, and exit the address setting through the corresponding function key "Done" at the bottom of the screen.

⑭Baud rate: The baud rate of power supply is 4800, 9600, 14400, 19200, 38400, 56000, 57600 and 115200 respectively.

Instruction:

⑮Protocol version: the type of power communication protocol is set, and the communication protocol can be selected from SCPI and MODBUS.

⑯Terminator: Set the power communication terminator. The communication terminator includes CR, LF, CRLF and LFCR.

⑰Reset: power reset, with options of "OK" and "Cancel", where "OK" means to reset the power supply and "Cancel" means not to reset the power supply. The power reset setting is to initialize the Menu function of the power supply at the factory.

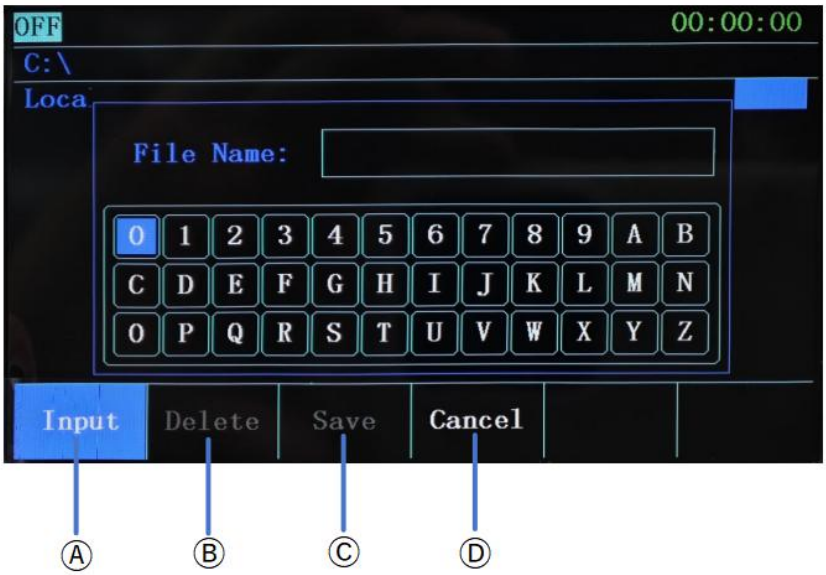
After initialization, the settings are as follows:

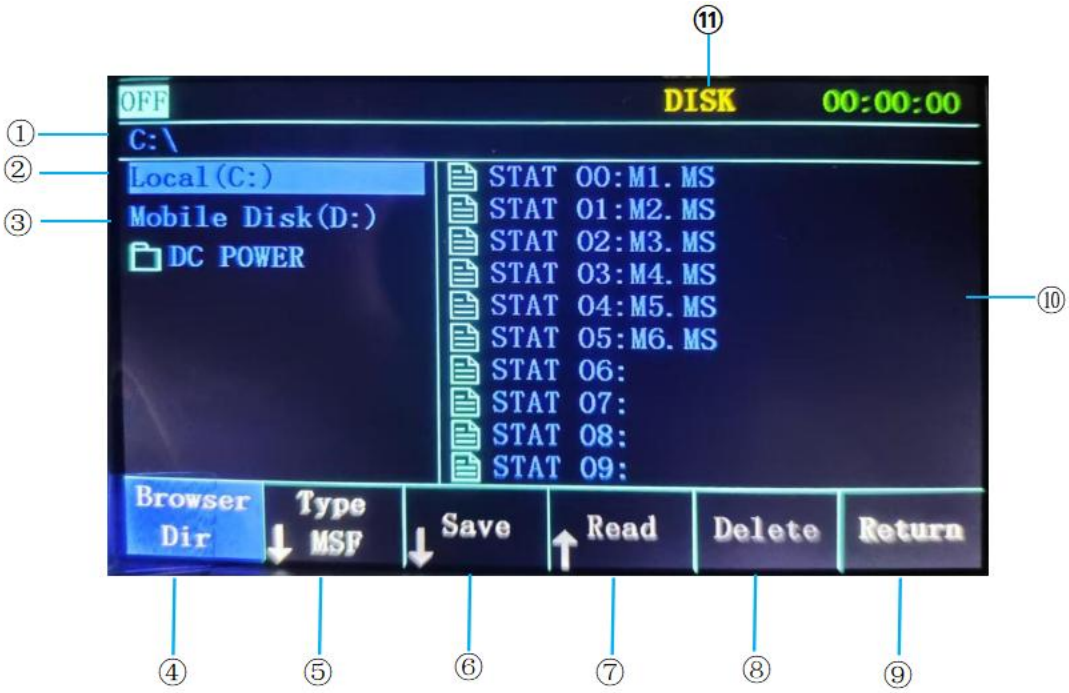
3.11 Esc (Cancel/Exit)

Esc (Cancel/Exit) key. Press Esc in the menu setting interface to exit the current window and return to the main interface. Press Esc in the parameter setting interface to cancel the current operation.

3.12 Store function

File storage and call of common voltage and current parameters of power supply, List function parameters and Battery function parameters.





①. c: \: Current disk location.
②. local disk (C): the storage disk inside the power supply.
③Removable disk (D): A storage disk outside the power supply.

④Browse Disk: used to select the disk to browse and the files in the disk, and enter the disk for browsing through the function key corresponding to "Browse Disk" at the bottom of the screen. When you have inserted a mobile disk (D), you can select the disk to browse by turning the knob left and right after entering the disk for browsing, and press the knob to enter the disk for browsing after selecting it. After entering the disk for browsing, you can save and read (call) the preset files by turning the knob left and right.

⑤. Type: used to select the file type stored or called in the disk. Select "File Type" by the corresponding function key at the bottom of the screen, and the file types are: *.MSF, *.MBF, *.MTF, *.CSV and other formats.

*.MSF: Storage file for voltage and current parameters commonly used in power supply.

*.MBF: It is the storage file of the Battery function parameter of the power supply.

*.MTF: Storage file for parameters of power supply "List operation".

*.CSV: a storage file for the parameters of the power supply "List operation". This file can only be saved and read when the mobile disk (D) is inserted.

⑥Save: file storage of commonly used voltage and current parameters of power supply, List function parameters and Battery function parameters. When "*.CSV" is selected, the mobile disk (D) is not inserted, and the save menu is grayed out and unavailable. Edit the commonly used voltage and current parameters, List function parameters and Battery function parameters in advance, press the Store key to enter the "access interface", and then enter the browse disk (refer to the first paragraph of this section④Item), select the location where the file is saved, such as local disk (c) or mobile disk (d), turn the knob left and right to select the disk, press the knob to confirm and enter the file list (refer to item in this section). After entering the file list, you can move the cursor up and down by turning the knob to select the file group to be saved, and then press the function key () corresponding to "Save" at the bottom of the screen to save. (as shown below). When saving, we need to give a name to the file. Select the "Enter" () key at the bottom of the screen and enter a name for the file by turning the knob to select the numbers or letters on the keyboard. If there is no file name, the "Save" menu is grayed out. Select the correct file type when saving (refer to item in this section). If you don't save, you can cancel and exit the saving interface by pressing the "Cancel" key (). If the input is wrong, you can delete the wrong input through the "Delete" key ().

⑦. Read: file call of commonly used voltage and current parameters of power supply, List function parameters and Battery function parameters. Enter the browse disk (refer to the first paragraph of this section④Item), select the disk location of the file to be called, and press the knob to enter the file list (item in this section). After entering the file list, you can select the file to be called by turning the knob and moving the cursor up and down. After selecting the required file, you can call up the data through the function key corresponding to "Read" at the bottom of the screen and enter the corresponding test. When there are no files in the selected corresponding group, the "Read" menu is grayed out. When "*.CSV" is selected, the mobile disk (D) is not inserted, and the "Read" menu is grayed out and unavailable.

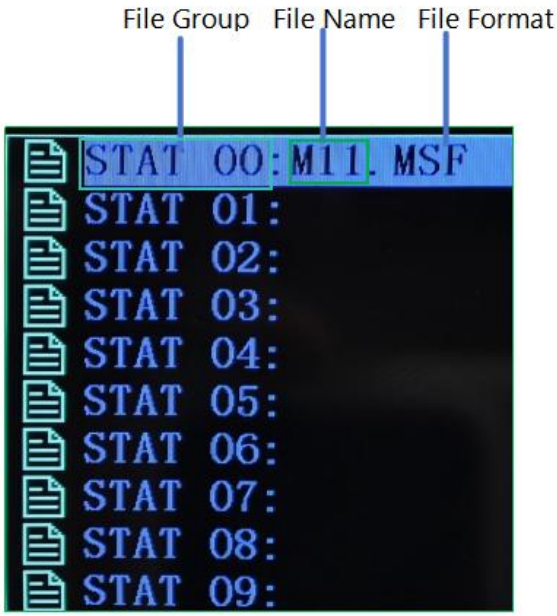
*.MSF: Call the commonly used voltage and current parameters of the power supply. After reading, the power supply returns to the basic operation interface, and the preset parameters such as voltage, current, OVP and OCP are called out.

*.MBF: refers to the call of the Battery function parameter of the power supply. After reading, the power supply returns to the Battery function interface and calls out the preset parameters.

*.MTF: refers to the call of the parameters of the power supply "List Operation". After reading, the power supply returns to the "List Settings" interface, and the preset parameters are called out. Press List to enter the "List Operation" to execute it.

*.CSV: It refers to the call of the parameters of "List Operation" of the power supply. This file can only be read when the mobile disk (D) is inserted. After reading, the power supply returns to the "List Operation" interface and calls out the preset parameters. Select the correct file type when reading the call (refer to item in this section).

⑧Delete: used to delete unused files. Enter the disk to browse through the function key corresponding to "Browse Disk" at the bottom of the screen. Turn the knob left and right to select the disk where the file to be deleted is located, then press the knob to enter the disk. Turn the knob left and right to select the file to be deleted, and delete the file through the function key corresponding to "Delete" at the bottom of the screen. At this time, the dialog box "Delete File?" will pop up. , press OK at the bottom of the screen to delete the file or Cancel to cancel this operation.
⑨. Return: After setting the parameters, return to the main interface at startup through the "Return" function key at the bottom of the screen.
⑩file list. Select the disk to browse and press the knob to browse the disk. At this time, all available files can be displayed here. By turning the knob left and right, the maximum number of groups is 100 (0-99), "STAT 00" means the first group, and "STAT 99" means the 100th group. The middle of the file name is the name of this file, which is defined by the customer, followed by the file format, which is divided into: *.MSF, *.MBF, *.MTF, *.CSV and other formats (as shown below).
⑪DISK: prompt for moving the DISK. When the "disk" is lit, it means that the mobile disk has been inserted.

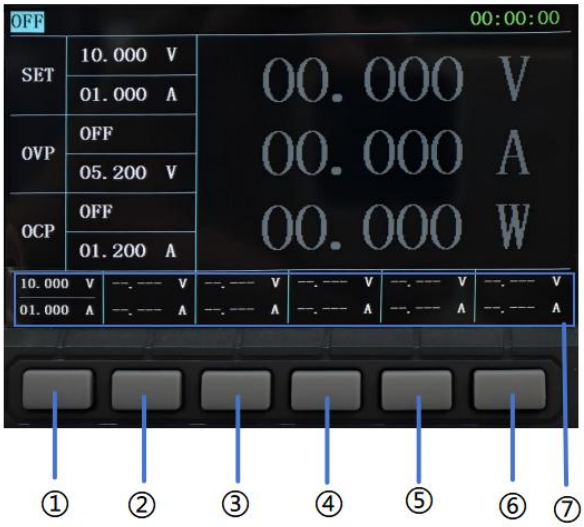


3.13 Hol Key function

The power supply can store some commonly used parameters in six groups of non-volatile memories for users to conveniently and quickly call out.

3.13.1 Preservation of Holkey function

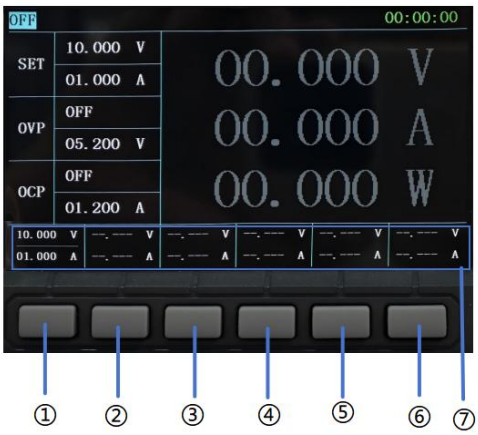
Set the parameters such as voltage and current to be saved in advance, press the HotKey key to enter the hotkey function (the screen is shown below), save the preset parameters such as voltage and current by long pressing the corresponding shortcut function key at the bottom of the screen, and press the HotKey key again to exit the hotkey function.



① Hotkey 1
② Hotkey 2
③ Hotkey 3
④ Hotkey 4
⑤ Hotkey 5
⑥ Hotkey 6
⑦ Display of voltage and current parameters corresponding to hotkeys

3.13.2 Call of Holkey function

Press HotKey key in the standby main interface to enter the hotkey function (the screen is shown below), call preset parameters such as voltage and current through the corresponding shortcut function key at the



① Hotkey 1
② Hotkey 2
③ Hotkey 3
④ Hotey 4
⑤ Hotkey 5
⑥ Hotkey 6
⑦ Display of voltage and current parameters corresponding to hotkeys

bottom of the screen, and press HotKey key again to exit the hotkey function.

3.14 Lock (keyboard lock) function

Press Lock to LOCK the power button. When the button is locked, the "Lock" indication in the function display bar at the top of the screen will be lit. Press Lock again to unlock the button.

Chapter IV Technical Specifications

4.1 Main technical specifications

MPS-400 series technical specification table

Model		Voltage	Current	Power
MPS-401		0~15V	0~60A	900W
MPS-402		0~20V	0~30A	600W
MPS-403		0~20V	0~45A	900W
MPS-404		0~30V	0~20A	600W
MPS-405		0~30V	0~30A	900W
MPS-406		0~60V	0~10A	600W
MPS-407		0~60V	0~15A	900W
MPS-408		0~80V	0~6A	480W
MPS-409		0~80V	0~11A	880W
MPS-410		0~150V	0~3A	450W
MPS-411		0~150V	0~6A	900W
Rated input voltage		AC 220V±10%		
Transform Mode		Switch power supply		
Load regulation rate	Voltage	≤0.1%+20mV		
	Current	≤0.1%+5mA		
Line regulation rate	Voltage	≤0.1%+20mV		
	Current	≤0.1%+5mA		
Setting resolution	Voltage	1mV		
	Current	1mA		
Setting accuracy	Voltage	≤0.1%+10mV		
	Current	≤0.3%+10mA		
Read back resolution	Voltage	1mV		
	Current	1mA		
Read back accuracy	Voltage	≤0.1%+10mV		
	Current	≤0.3%+10mA		
Temperature	Operating	0 to 40℃≤85 R.H.		
	Storage	-15 to 70℃≤85 R.H.		
Size (W * H * D)	mm	230*110*408		
Net weight	kg	4.5		

WPS-400 series technical specification table (2);

Model		Voltage	Current	Power
WPS-401		0~30V	0~40A	400W
WPS-402		0~30V	0~60A	900W
WPS-403		0~40V	0~40A	400W
WPS-404		0~40V	0~40A	900W
WPS-405		0~60V	0~20A	400W
WPS-406		0~60V	0~30A	900W
WPS-407		0~80V	0~20A	400W
WPS-408		0~80V	0~20A	900W
WPS-409		0~150V	0~10A	400W
WPS-410		0~150V	0~15A	900W
Rated input voltage		AC 220V±10%		
Transform Mode		Switch power supply		
Load regulation rate	Voltage	≤0.1%+20mV		
	Current	≤0.1%+5mA		
Line regulation rate	Voltage	≤0.1%+20mV		
	Current	≤0.1%+5mA		
Setting resolution	Voltage	1mV		
	Current	1mA		
Setting accuracy	Voltage	≤0.1%+10mV		
	Current	≤0.3%+10mA		
Read back resolution	Voltage	1mV		
	Current	1mA		
Read back accuracy	Voltage	≤0.1%+10mV		
	Current	≤0.3%+10mA		
Temperature	Operating	0 to 40℃≤85 R.H.		
Temperature	Storage	-15 to 70℃≤85 R.H.		
Size (W * H * D)	mm	230*110*408		
Net weight	kg	4.5		

4.2 Supplementary Features

Recommended calibration frequency: 1 year/1 time.
Heat dissipation mode: forced air cooling
Operating environment temperature: 0 to 40℃
Storage environment temperature: -20 to 70℃
Use environment: indoor use design, pollution level 2, maximum humidity 80%

Chapter 5 Communication between Power Supply and PC

Communication between power supply and PC

The power supply comes standard with three communication interfaces: RS232, USB and RS485, and users can choose any one to communicate with the computer.

Communication settings:

Before communication, you should first match the power supply with the following parameters of the PC.
1. Baud rate: 9600(4800, 9600, 14400, 19200, 38400, 56000, 57600, 115200). You can enter the system menu through the panel and set the communication baud rate.
2. Consultation: Set the type of power communication protocol, and the communication protocol can be selected from SCPI and MODBUS.
3. Terminator: set the power communication terminator. The communication terminator includes LF, CR, CRLF and LFCR (the terminator must be correct in SCPI protocol).
4. Address: set the power communication address, ranging from 1 to 254 (the address must be correct in MODBUS protocol).

Data bit: 8
Stop bit: 1
Check: NONE(8 data bits have no check)

Start Bit	8 Data Bits	Parity=None	Stop Bit

Attachment FAQ

1. Cannot boot, (please refer to the exception handling of Quick Start).
2. Can't communicate with PC. Enter the menu to check the configuration parameters of communication, such as address, baud rate, instruction type and terminator (please refer to the configuration parameters of menu function.)
3. No output:

1. Check whether the output is turned on by pressing the On/Off key (see 3.4 Output On/Off Operation).
2. Check that the power supply or current is set to 0.

Warranty Card

What the warranty covered:

If the machine break down due to its defectiveness, MATRIX will provide free maintenance during warranty period. If the machine break down due to wrong operation or carelessness, then Matrix provide paid service within warranty period.

How long does this warranty last:

This warranty lasts for 3 years from the date of original purchase of all MATRIX branded products.

Who is covered:

This warranty covers only the original purchaser of this product. This warranty is not transferable to subsequent owners or purchasers of this product.

What do customers need to do to get repairs/service under the warranty policy?

If the machine get problem, please contact our local distributor. If you cannot find the local distributor, you can contact us directly, our email is service@szmatrix.com, our telephone No. is 0086 755 2836 4276.

What information do customers need to supply?

Model No.	
Serial No.	
Problem description	
Picture	
Video if necessary	